

PROJECT



Sentiment Analysis of Tweets that Express Political Leanings

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ABSTRACT

Social media platforms, especially Twitter, have changed significantly. They are now key channels for understanding public opinion, political discussions, and societal trends. The fast and constant creation of content by users makes Twitter a useful source for analyzing political feelings and predicting how people will react to major political events. With Natural Language Processing (NLP) improving, Large Language Models (LLMs) can effectively interpret complex political language, identify ideological patterns, and assess changes in public mood.

This paper looks at how LLM-based text analysis can predict political leaning and possible election trends from posts shared on Twitter. It examines the methods used in previous studies, including machine learning, deep learning, and mixed approaches. The review also points out how modern NLP tools like BERT, GPT, and other transformer models improve classification accuracy and understanding of context. Additionally, the study reviews how preprocessing methods, feature extraction techniques, and sentiment lexicons enhance model performance.

The main goal of this research is to provide a clear overview of current methods used in political sentiment analysis, identify limitations, and suggest areas for future research. By combining recent findings, this work seeks to aid future projects on LLM-driven sentiment and ideology detection for political analysis and election trend prediction.

INTRODUCTION

Social media sites, platforms such as Twitter have evolved into essential channels for disseminating information voicing opinions and influencing public discourse. Their influence, on communication continues to expand as individuals, political entities and organizations use these platforms more frequently to engage with each other. This immediate and dynamic mode of communication has transformed participation. Today users are able to express their perspectives respond promptly to occurrences. Engage in communal decision-making processes.

By 2025 Twitter, boasting than 525 million active users each month has emerged as a crucial arena for political discourse. Its open architecture, communication and worldwide accessibility make it particularly valuable for analyzing political conduct. Earlier research shows that social media information can predict election outcomes, pinpoint factions, uncover trends in political dialogue and illuminate exchanges, among political leaders. Additionally, social media campaigns and focused initiatives have been demonstrated to influence engagement and voter participation.

Nevertheless, the extensive use of media has resulted in the rapid dissemination of misinformation, biased perspectives and false information. These issues have the potential to distort opinion and threaten political stability. The presence of news, propaganda and emotionally manipulative political messages highlights the necessity, for tools that can detect sentiment, political orientations and the legitimacy of online conversations.

Discussions on politics frequently revolve around global matters. For instance, debates concerning the Palestinian conflict evoke intense emotions and capture worldwide interest. Such subjects regularly impact ties, financial decisions, demonstrations and cultural activities. Politicians occasionally employ -political material deliberately to sway public sentiment regarding nations or international affairs. The employment of hashtags the rapid dissemination of information and the audience's reaction to intense material all demonstrate Twitter's strength, in molding political stories.

Grasping the tone and political background of messages is particularly crucial in debates since replies may alter or completely flip the meaning of the initial posts. Recent research has introduced machine-learning models designed to identify "opinion inversion," a phenomenon where the sentiment in a quoted tweet contrasts with that of the original. Focusing on conversations, about the Palestinian conflict scholars have examined how quote tweets either contradict or reinforce the sentiment expressed in the source text. These research findings reveal tactics in political messaging such, as introducing unrelated topics to shift the mood or course of public debates.

As polarization increasingly affects digital political settings, automated systems for analyzing public sentiment and ideological alignment have become crucial. With the development of Large Language Models (LLMs), analyzing Twitter text can now identify political leanings and predict broader political trends, including possible changes in election outcomes. This research explores machine-learning and LLM-based methods for predicting political orientation and election-related trends using Twitter data.

Background

Sentiment analysis has been shown to be an important tool in almost every research . As such, a large number of analysis have been published on the topic. Sentiment analysis has been applied to social media and tweeter in an attempt to capture political system; some of these results have shown that social media and online judgment content can reliably predict certain political events.

Our research focused on *Twitter*, an extended a big platform whose users share short messages called *tweets* that can include links to other websites, videos or pictures. Given that politics is an important source of views, it is an active topic on this social internet network. The application of sentiment analysis technological and the Twitter and social media is a popular research topic today, and many studies have attempted to use these technologies.

This section details reviews the previous studies on the topic of sentiment analysis of political messaging. We first explore the landscape of political sentiment analysis technology that rely on Twitter and then discuss the main methods used to extract sentiment information from written content.

Related Works

Research on political sentiment analysis using Twitter and other social media platforms has evolved significantly over the past decade. Early studies focused on traditional machine-learning techniques such as decision trees and k-nearest neighbors to classify political opinions expressed by Italian Twitter users. Other researchers examined political trends in the Venezuelan parliamentary elections of 2015 using unsupervised sentiment analysis methods combined with linguistic tools such as the Linguistic Inquiry and Word Count (LIWC) to identify emotional and psychological patterns in tweets.

Deep learning models have also been applied to political prediction tasks. For instance, long short-term memory (LSTM) networks were used to infer the outcome of the 2019 Indian general elections, and their performance was compared with conventional machine-learning algorithms. Although several studies reported promising results, other researchers argued that sentiment polarity alone—positive or negative—was not a reliable signal for predicting voter preference. An example of this limitation was observed during the 2016 U.S. presidential election, where sentiment extracted from tweets did not consistently correlate with support for a given candidate.

Despite these limitations, other works demonstrated that social media sentiment can still provide relevant insights into political outcomes. Using a lexicon-based approach, researchers analyzed polarity and bias in tweets during the 2016 U.S. presidential election and concluded

that public sentiment did reflect patterns associated with electoral preferences. Similar studies adopted four standard stages—data retrieval, preprocessing, analysis, and visualization—to evaluate political discourse. These included the use of word clouds, sentiment dictionaries, and lexical resources to interpret frequently occurring terms. Several authors also emphasized the importance of high-quality sentiment lexicons, noting that lexicons translated from one language to another often fail to capture context-specific sentiment accurately.

Further contributions analyzed sentiment during the Spanish presidential elections of 2015 and 2016 using NLP and text-mining techniques based on the LinguaKit dictionary, categorizing tweets into positive, negative, or neutral classes. Other researchers utilized emoticons as sentiment cues, assigning polarity scores based on emoji usage within political tweets.

A different strand of research explored the influence of malicious or automated accounts on political sentiment formation. These studies employed semi-supervised methods using word embeddings to classify political hashtags, identify interest groups, and detect coordinated manipulation or “thread hijacking.” Several works used convolutional neural networks trained on the Sentiment140 dataset to create large-scale political sentiment classifiers. In related research, word embeddings were learned from a corpus of four million tweets gathered during the 2016 Brexit referendum to capture semantic relationships in political language. Additional studies estimated political polarization using advanced neural methods such as Iterative Opinion Mining using Neural Networks (IOM-NN), while others proposed cloud-based frameworks to analyze political sentiment on scale.

Important related contributions also include the development of corpora such as the Corpus of Spanish Tweets (COST), designed specifically for political sentiment classification. Other studies demonstrated that the effectiveness of sentiment analysis strongly depends on the quality of lexicons, prompting researchers to build new dictionaries for political events such as the 2015 Greek referendum, incorporating techniques like sarcasm correction. Additional work introduced political sentiment indices to track public attitudes during events such as the 2015 U.K. elections. Finally, a dedicated sentiment analysis model for social and political events was proposed to measure public interest and reactions, highlighting the utility of lexicon-based systems in understanding societal opinions.

DATASET

An organized and detailed data set is crucial for any research focused on prediction especially in the realm of political sentiment analysis. The precision and dependability of machine-learning or LLM-based models largely rely on the caliber, variety and volume of the data utilized. Social media networks offer a collection of openly accessible political material making them important resources, for examining public opinions, ideological leanings and developing political movements.

In this study, we utilize publicly available datasets from Hugging Face, specifically the Political Bias and Political Leaning datasets. These datasets contain large collections of political text, primarily drawn from Twitter and other online platforms. They include annotated labels indicating sentiment—Positive, Negative, and Neutral—as well as political orientation categories. Such annotations enable systematic evaluation of how user-generated content reflects ideological positioning and public reactions to political events.

The full dataset contains approximately 128,484 data instances. For experimental purposes, a subset of 10,000 samples was selected to ensure manageable computation while maintaining statistical representativeness. Using a larger dataset helps improve the generalizability of the results, whereas very small datasets may lead to biased or unreliable conclusions. The chosen subset preserves diversity across political viewpoints and sentiment categories.

To evaluate model performance across different prompting strategies, we employ three approaches: zero-shot, one-shot, and few-shot prompting. This allows us to compare how LLMs interpret political sentiment and leaning with varying levels of contextual guidance. By applying these prompting methods to a substantial dataset, the study provides meaningful insights into the ability of LLMs to analyze political discourse and predict potential trends. Annotation was performed using the following three labels:

- pos: Tweets which carry positive sentiment towards the topic .
- neg: Tweets which carry negative sentiment towards the topic .
- neu: Tweets which do not carry any sentiment towards the topic.

Liberal (Positive) Tweet	9992
Conservative(Negative) tweet	4015
Neutral tweet	114477
Total Number	

Methodology:

To assess model effectiveness across prompting methods we utilize three techniques, one of which is zero-shot prompting leveraging transformer-based model inference. This method allows us to gauge a model's grasp of political sentiment and ideological bias without relying on task-tailored fine-tuning or annotated samples. By contrasting results produced under prompting scenarios we examine how large language models (LLMs) process political narratives, with different degrees of contextual cues. These prompting methods are applied to a substantial and diverse dataset to ensure robustness and generalizability of the results. The evaluation provides meaningful insights into the effectiveness of LLMs in analyzing political discourse, identifying sentiment polarity, and predicting potential ideological trends. Additionally, multiple transformer-based architectures are utilized in this study to systematically compare their performance, allowing us to identify which model demonstrates superior accuracy, consistency, and reliability in political bias and sentiment classification tasks.

Preprocessing:

After loading the dataset, we have separated the text and its corresponding label so that we can use the text samples using into the model for inference.

Model Processing:

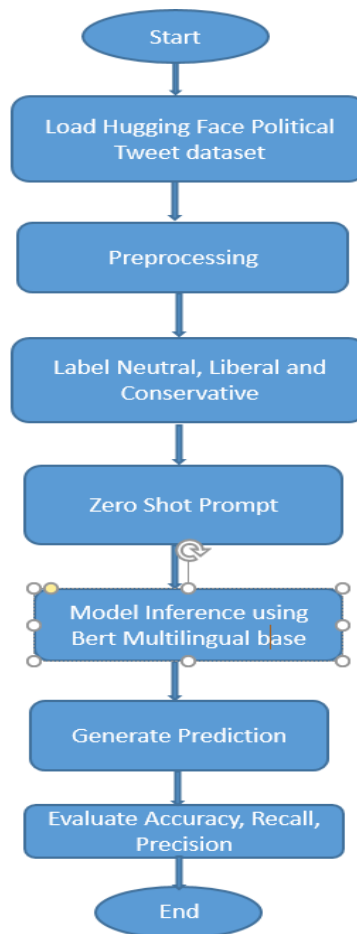
We have 3 Transformer model for the comparison.

- DistilRoBERTa
- Twitter-RoBERTa
- BERT Multilingual Base (Uncased)

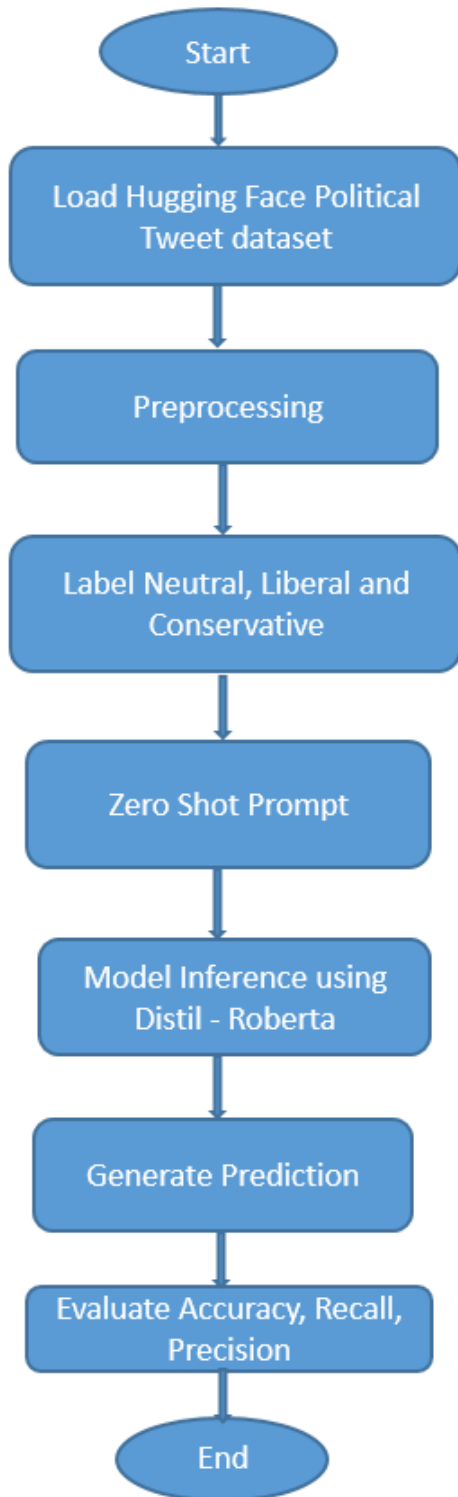
DistilRoBERTa is a distilled (compressed) version of RoBERTa, created by Hugging Face. It's a smaller, faster model .

Twitter-RoBERTa (officially: cardiffnlp/twitter-roberta-base-sentiment) is a domain-specific version of RoBERTa that has been fine-tuned specifically on Twitter/social media data for sentiment analysis and other Twitter-related NLP tasks.

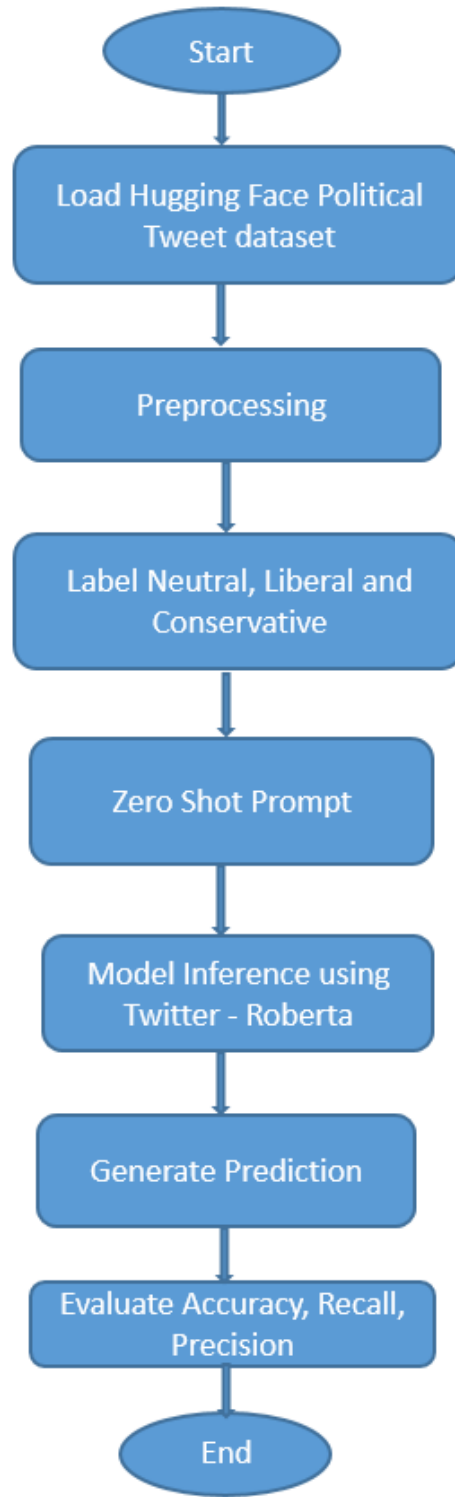
We have used implement the above model for 10,000 samples of out of total 128,484 samples from the dataset. First, we segregate the text and label in different variable, the apply model inference on the text to predict the label by the model individually. After that, Accuracy, Precisoin, F1-score have been claculated by comparing the true labels and predicted labels for each models



(a) Bert Multilingual base



(b) Distil Roberta



(c) Twitter Roberta

Fig-3: Simplified Flow diagram

Evaluation Metrics:

When building machine learning models, it's important to understand how well they perform. Evaluation metrics help us to measure the effectiveness of our models. Whether we are solving a classification problem, predicting continuous values or clustering data, selecting the right evaluation metric allows us to assess how well the model meets our goals. In this article, we will see commonly used evaluation metrics and discuss how to choose the right metric for our model.

Classification Metrics (Binary & Multiclass)

Let's define for binary classification first:

TP = True Positives

TN = True Negatives

FP = False Positives

FN = False Negatives

Accuracy:

Proportion of correctly predicted samples. For imbalance datasets it can be misleading. Accuracy is good but it gives a False Positive sense of achieving high accuracy. The problem arises due to the possibility of misclassification of minor class samples being very high.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision:

Fraction of predicted positives that are actually positive. It's useful when the cost of false positives is high such as in medical diagnoses where predicting a disease when it's not present can have serious consequences.

$$Precision = \frac{TP}{TP + FP}$$

Recall:

Measures how well the model detects positive cases. Sensitivity is also termed as Recall or True Positive Rate (TPR) which measures how well true positives are identified by a test or model. It depicts among all the positive cases how many were correctly identified as positive by the model.

$$Recall = \frac{TP}{TP + FN}$$

F1-Score:

Harmonic mean of precision and recall. It is useful when we need a balance between precision and recall as it combines both into a single number. A high F1 score means the model performs well on both metrics. Its range is [0,1].

Lower recall and higher precision gives us great accuracy but then it misses a large number of instances. More the F1 score better will be performance. It can be expressed mathematically in this way:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Distil-Roberta using 10000 samples –

Accuracy: 86%

F1 Score: 89%

Precision: 91% (confident predictions)

Recall: 86% (able to capture majority class cases)

Balanced Performance: No severe bias toward any class

Accuracy: 52%

F1 Score: 65%

Precision: 93% (confident predictions)

Recall: 52% (able to capture majority class cases)

Result Analysis and Summary:

In our result analysis we have observed that the accuracy is high and steady, deviation is not much. F1-score, Precision and Recall is also good.

Data Count	Accuracy	F1-Score	Precision	Recall
100	89	94	100	89
500	83	90	99	83
1000	82	90	99	82
2000	84	89	94	84
3000	84	89	94	84
4000	83	87	92	83
5000	82	86	91	82
6000	82	87	91	82
7000	83	86	90	83
8000	83	87	90	83
9000	85	88	91	85
10000	86	89	91	86

Table-01: Result Summary using DistilRoberta

But in twitter Roberta the accuracy is poor and also the f1-score, precision and recall is also same way poor.

Sample	Accuracy	F1-Score	Precision	Recall
100	39	56	100	39
500	54	70	98	63
1000	63	77	99	63
2000	52	67	95	52
3000	43	59	94	43
4000	41	57	92	41
5000	43	58	91	43
6000	44	60	92	44
7000	46	61	90	46
8000	46	61	91	46
9000	49	64	92	49
10000	52	66	93	52

Table-02: Result Summary Using Twitter Roberta

Sample	Accuracy(%)	F1-Score (%)	Precision(%)	Recall(%)
100	12	21	100	12
500	20	33	99	20
1000	20	33	100	20
2000	15	24	96	15
3000	15	24	96	15
4000	16	24	92	16
5000	17	26	89	17
6000	17	25	89	17
7000	18	27	89	18
8000	18	28	90	18
9000	15	23	90	15
10000	14	22	90	14

Table-03: Result Summary Using Bert Multilingual base

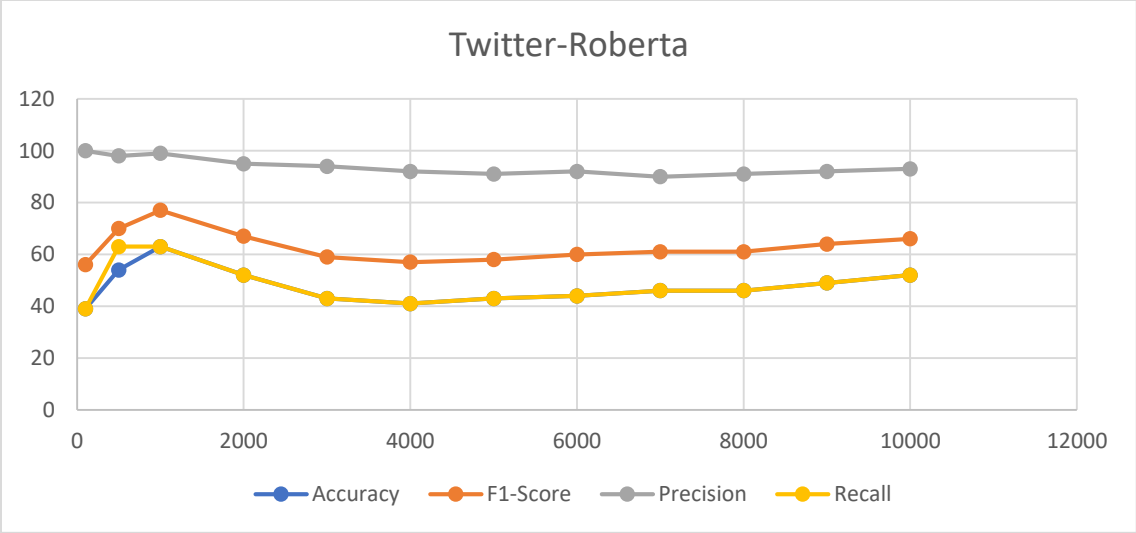


Fig:04 Comparison using Twitter Roberta

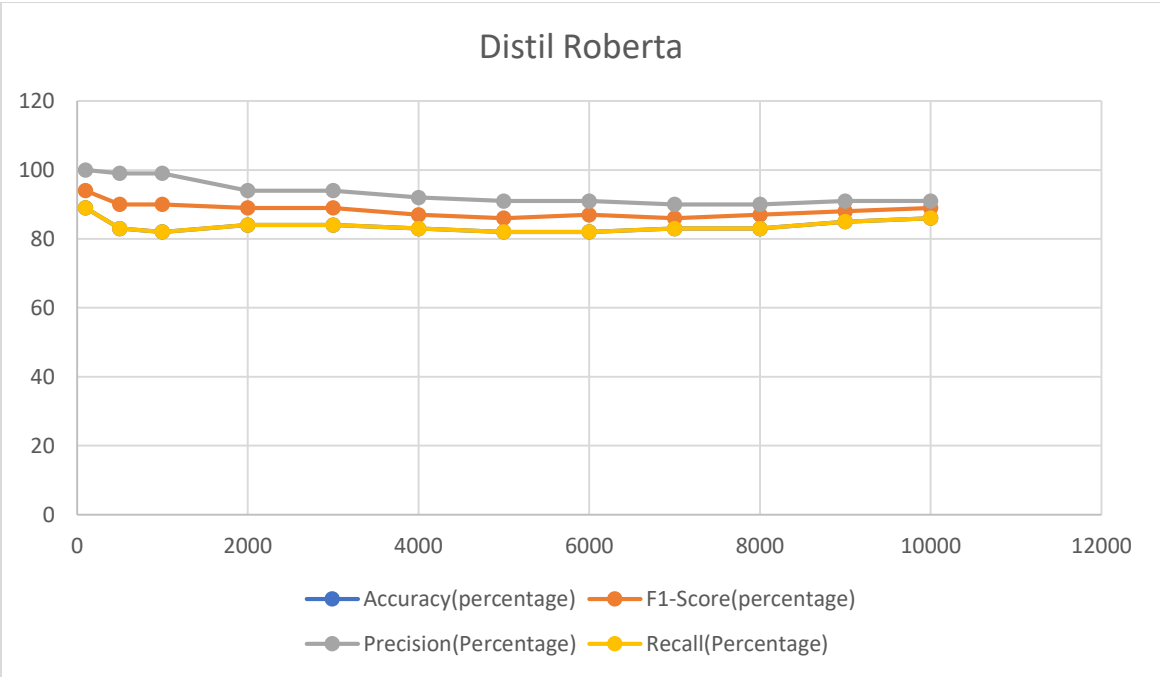


Fig: 05 Comparison using DistilRoberta

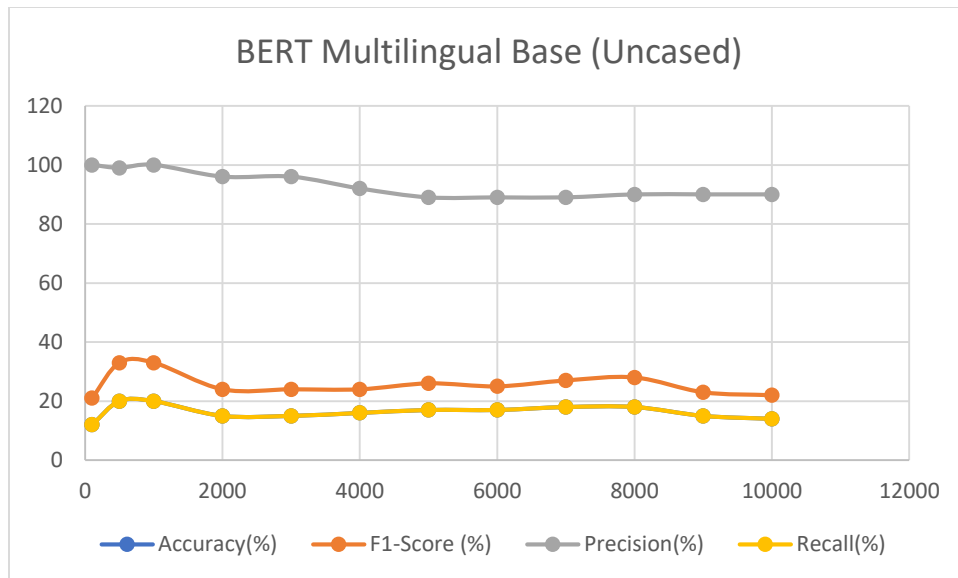


Fig: 06 Comparison using Bert Multilingual base

Future Work

If we can get to success this research paper then we can apply our country. We know our country is very small and over populated. But we have a lot of resource. If any political government does their working system harmful then we can apply this project our Bangladeshi people. In this project we are doing only four model but we can apply it more. When our prediction is accurately working then we can use widely. Also political people use this system for understanding their situation. We also comparison with LLM models like GP14, Claude etc.

We also use balanced dataset with almost equal partition of all types of samples.

Discussion and Conclusion

The results of this research highlight the increasing significance of analyzing sentiment and political orientation to grasp sentiment throughout election times. Twitter continues to be a space where political stories change swiftly enabling immediate tracking of public responses ideological splits and new voting tendencies. The evaluation shows that Large Language Models

(LLMs) utilized via zero-shot one-shot and few-shot prompting methods are capable of identifying relevant patterns in politically charged social media posts.

Unlike studies that mainly depended on conventional machine-learning methods this technique utilizes the contextual understanding abilities of advanced LLMs to assess sentiment with greater precision and detect hidden political signals in text. Earlier research frequently faced difficulties because of the vagueness and casual style of media language while LLMs offer improved flexibility and resilience across varied political subjects and language patterns. Employing an varied dataset, from Hugging Face additionally boosted the trustworthiness of the obtained findings.

Nonetheless numerous obstacles persist. Social media information includes noise, prejudice, information orchestrated manipulation and emotionally intense material, each capable of affecting algorithmic forecasts. Although LLMs are powerful they might still fail to grasp sarcasm, subtle political allusions or unique phrases. Guaranteeing equity minimizing model bias and confirming predictions, through electoral results continue to be active areas of investigation.

In summary this research shows that contemporary LLM-based sentiment analysis offers benefits for political studies and can aid decision-making for political groups, policymakers and analysts. Through the combination of Twitter data and sophisticated NLP methods our study enhances the comprehension of online political conversations and lays the groundwork, for future tools designed to predict election trends and evaluate public opinion with greater precision.

Future research may explore fine-tuning domain-specific models, incorporating multimodal political data, improving misinformation detection, and enhancing temporal prediction of political events. Such advancements can further strengthen the applicability of LLM-driven sentiment analysis in real-world political environments.